

BOILER STEAM CONSUMPTION AUDIT AND FIELD INSTRUCTIONS

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Source reviewed: Aqua Analytics Boiler Report, August 12, 2026

Known from report	Water meter increased 88,802.7 gal from 7/23/2026 to 8/12/2026, averaging 4,440.14 gal/day.
Equivalent water mass	About 37,031 lb/day of metered water, or about 1,543 lb/hr averaged over 24 hours.
Critical unknown	The report does not identify where the water meter is installed. Meter location determines whether this is makeup water or total boiler feedwater.
Best current estimate	If the meter is total boiler feedwater, a chloride-based cycles estimate suggests roughly 25,900 lb/day of steam, about 1,080 lb/hr averaged over 24 hours. This is an estimate, not a verified plant steam total.

Purpose of this audit

This packet explains what can and cannot be calculated from the existing boiler report, identifies the missing field information needed to determine actual steam consumption, and provides a practical collection worksheet for closing the boiler water and steam mass balance.

Bottom line

- The report is useful enough to establish a measured water-use baseline of 4,440.14 gal/day.
- That water-use number is NOT automatically the same as steam consumption.
- The single most important next step is to physically identify the water meter and determine whether it measures city/makeup water, softened makeup water, or total boiler feedwater.
- Once meter location is confirmed, condensate return and blowdown information can be added to calculate or tightly estimate steam production and treatment efficiency.

FIELD PRIORITY: FIND THE WATER METER AND TRACE BOTH SIDES OF THE PIPE.

1. What the Aqua Analytics report actually tells us

Water meter and feedwater temperature

Report item	Previous	Current / Difference	Interpretation
Water meter	6,165,009 gal	6,253,811.7 gal / +88,802.7 gal	20-day average = 4,440.14 gal/day
Feedwater temperature	184 F	150 F / -34 F	Report recommendation: feedwater should be 180 F

Water-use conversions from the report

Measured Water Baseline

$$4,440.14 \text{ gal/day} \times 8.34 \text{ lb/gal} = 37,030.8 \text{ lb water/day}$$

$$37,030.8 \text{ lb/day} / 24 = 1,542.95 \text{ lb water/hr}$$

$$4,440.14 \text{ gal/day} / 24 = 185.0 \text{ gal/hr}$$

Important correction: the average metered flow is about 185 gal/hr across 24 hours. The mass equivalent is about 1,543 lb/hr.

Water chemistry findings that matter to the steam calculation

Parameter	Feedwater	Boiler	Report target / comment
Conductivity	1,335 umhos	4,547 umhos	Boiler target 2,500-3,500; feedwater conductivity called abnormal
Chloride	30 ppm stated in comments	100 ppm	Boiler target shown 30-65 ppm
pH	10.75	11.64	Feedwater 7.5-10.5; boiler 10.5-11.8
P Alkalinity	150 ppm	560 ppm	Feedwater 4-20; boiler 300-700
Sulfite	36 ppm	36 ppm	Boiler target shown 60-80 ppm
Feedwater hardness	0 ppm	0.5 ppm	Softener and boiler hardness are low, but salt carryover was noted

Report comments affecting reliability

- Boiler described as overcycled with insufficient blowdown, creating sludge and carry-over risk.
- Feedwater sample had abnormally high conductivity, foaming, and a strong chemical smell.
- Technician suspected excessive condensate treatment or chemical returning through a steam trap; C41 pump was reduced from 12 seconds to 6 seconds.
- Softener rinse time should be increased because chloride/salt is leaving the softener and reaching feedwater.
- Feedwater should be 180 F; the current reading was 150 F.

2. Can this report tell us actual steam consumption?

Yes - but only after the physical location and function of the listed water meter are confirmed. The same 4,440.14 gal/day reading means very different things depending on where the meter sits in the system.

Scenario A - meter measures TOTAL BOILER FEEDWATER

If the meter is downstream of the feedwater/condensate tank and measures all water pumped into the boiler, then feedwater equals steam production plus blowdown. In that case, boiler cycles can be used to estimate the blowdown fraction.

Chloride-Based Estimate

$$\text{Estimated cycles} = \text{Boiler chloride} / \text{Feedwater chloride} = 100 / 30 = 3.33$$

$$\text{Approx. blowdown fraction} = 1 / \text{cycles} = 1 / 3.33 = 30\%$$

$$\text{Approx. steam fraction} = 70\% \text{ of feedwater}$$

$$37,030.8 \text{ lb/day} \times 70\% = \text{about } 25,922 \text{ lb steam/day}$$

$$25,922 \text{ lb/day} / 24 = \text{about } 1,080 \text{ lb steam/hr average}$$

Working estimate if Scenario A is confirmed: approximately 25,900 lb/day of steam, or 1,080 lb/hr averaged over 24 hours.

Why conductivity should NOT be the primary basis right now

A conductivity ratio would give about 3.41 cycles (4,547 / 1,335). However, the report specifically says feedwater conductivity is abnormal and likely influenced by chemical contamination. That makes conductivity less trustworthy for a clean cycles-of-concentration estimate until the feedwater chemistry problem is corrected.

Scenario B - meter measures MAKEUP WATER only

If the meter is on city water or softened makeup water feeding the condensate/feedwater tank, then the meter measures replacement water, not total steam production. Condensate returning from the plant can be reused many times without passing through the makeup meter.

Makeup-Water Mass Balance

$$\text{Makeup water} = \text{unreturned condensate/steam} + \text{blowdown} + \text{miscellaneous losses}$$

Steam production may be much higher than the makeup-water reading when condensate return is high.

Under Scenario B, the 4,440.14 gal/day value is still very useful because it quantifies makeup demand and system losses, but it cannot by itself be labeled as plant steam consumption.

Scenario C - meter is somewhere else

If the meter serves more than the boiler system, or is installed before a branch that supplies other equipment, the report reading must be separated from non-boiler users before it can be used for a steam balance.

3. Boiler efficiency and chemical-treatment audit findings

A. Feedwater temperature is low

The report shows feedwater at 150 F and states it should be 180 F. The previous reading was 184 F. Lower feedwater temperature increases the sensible heat the boiler must add before vaporization, increasing fuel required for the same steam output.

B. Blowdown appears insufficient

The report explicitly describes the boiler as overcycled and recommends increasing blowdown. Boiler conductivity of 4,547 umhos is above the shown 2,500-3,500 target, while boiler chloride of 100 ppm is above the shown 30-65 ppm target. Excessive concentration increases carry-over and deposit risk.

C. Feedwater chemistry is abnormal

The technician observed foaming and a strong chemical smell in the feedwater sample. The report states that the measured P alkalinity, orthophosphate, and sulfite should normally correspond to only about 300-400 conductivity, yet measured feedwater conductivity was 1,335. This suggests contamination or excessive chemical return/dosing rather than normal dissolved solids alone.

D. Softener salt carryover is present

The report states that chloride/salt is leaving the softener and reaching the feedwater. Increasing softener rinse time was recommended. This matters because chloride is often used as a cycles indicator; a contaminated makeup/feedwater baseline can distort blowdown calculations.

E. Chemical feed should be normalized to production

Once actual steam production or total feedwater is known, chemical use can be tracked against production instead of only pump run time. Recommended operating metrics include:

- chemical gallons per day and ounces per hour;
- chemical per 1,000 gallons of makeup water;
- chemical per 1,000 lb of steam produced;
- makeup gallons per 1,000 lb of steam;
- blowdown percent of feedwater;
- condensate return percent;
- fuel per 1,000 lb of steam, when gas use is available.

What becomes possible after steam consumption is established

Metric	What it tells the operator
Steam lb/hr, lb/day, lb/month	Actual production load and trend
Makeup-water ratio	How much water is being lost instead of returned
Condensate return %	Direct indicator of heat and water recovery
Blowdown %	Whether water treatment is forcing excessive water/energy loss
Chemical per 1,000 lb steam	Whether treatment dosage tracks actual boiler work
Fuel per 1,000 lb steam	Overall boiler energy efficiency and cost

Do not adjust chemical feed solely from this packet. Use the treatment provider's limits and test results. This audit is intended to establish the water/steam balance needed for better dosing and efficiency decisions.

4. Field audit - meter identification and steam-balance worksheet

Complete this page at the boiler room. A photo of the meter plus enough piping on both sides to show where it comes from and where it goes is highly valuable.

Step 1 - identify the report water meter

Field	Observed / recorded value
Meter manufacturer/model	
Meter displayed total	
Units	US gal
Exact physical location	
Pipe entering meter comes from	
Pipe leaving meter goes to	
Any branches before/after meter	
Photo numbers/files	

Step 2 - classify the meter

- ☐ City/raw water before softener
- ☐ Softened makeup water to condensate/feedwater tank
- ☐ Total feedwater leaving tank/deaerator toward boiler
- ☐ Boiler blowdown meter
- ☐ Other: _____

4A. Field audit continued - operating data and condensate return

Step 3 - collect one normal operating shift of data

Reading	Start	End	Notes / units
Water meter total			
Shift start/end time			
Feedwater tank level			
Feedwater temperature			
Boiler conductivity			
Feedwater conductivity			
Boiler chloride			
Feedwater chloride			
Blowdown counter/time/flow if available			
Fuel meter / gas use if available			
Chemical pump totals or drum level			

Step 4 - condensate return information

Is there a condensate return meter? ☐ Yes ☐ No

If yes, meter location and reading: _____

Does the feedwater tank receive visible condensate return? ☐ Yes ☐ No

Are there known steam users that dump condensate to drain? _____

Known steam leaks / failed traps / open vents: _____

5. Calculation worksheet after meter location is confirmed

If meter = total boiler feedwater

Calculation	Formula	Result
Feedwater mass/day	Feedwater gal/day x 8.34 lb/gal	_____
Cycles of concentration	Boiler chloride / feedwater chloride	_____
Blowdown fraction	1 / cycles	_____
Steam fraction	1 - blowdown fraction	_____
Steam lb/day	Feedwater lb/day x steam fraction	_____
Steam lb/hr	Steam lb/day / operating hours	_____

If meter = makeup water

Do not label makeup-water flow as steam production. Use one of the following paths:

- Measure total boiler feedwater directly, then subtract blowdown.
- Measure condensate return and blowdown, then close the mass balance.
- If direct flow meters are unavailable, use a controlled timed test with tank-level change, pump capacity/runtime, or another verified plant measurement method.

Recommended final dashboard once data is verified

Daily KPI	Target / baseline	Actual
Steam production, lb/day	Establish baseline	_____
Makeup water, gal/day	Current report baseline 4,440	_____
Condensate return, %	Establish baseline	_____
Blowdown, %	Based on treatment limits	_____
Feedwater temp, F	Report says 180 F	_____
Chemical use per 1,000 lb steam	Establish baseline	_____
Fuel per 1,000 lb steam	Establish baseline	_____

Current conclusion

The report supports a measured boiler-system water baseline of 4,440.14 gal/day. It does not, by itself, prove total steam consumption. If the listed water meter is confirmed to measure total boiler feedwater, a chloride-based estimate gives approximately 25,900 lb/day of steam (about 1,080 lb/hr averaged across 24 hours). If it is a makeup-water meter, condensate return and/or total feedwater must be measured before total steam production can be determined.

NEXT ACTION: LOCATE, PHOTOGRAPH, AND TRACE THE WATER METER.